

Disabilities

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

Andrew Begel

Administrivia

- Homework 1 is due Monday, August 31, 11:00am via Gradescope.
- If you are not yet registered for this class, please come and see me after class.

Questions about Disability

- Definitions matter
- Is it a thing you are, or a thing you have?
- Can anyone else tell?
- How long have you been disabled and how long will it last?
- What functions does it impact?
- How did you get it?
- What are the odds that you get it?
- Do you have more than one?

Person-First vs. Identity-First Language

Person-first language



"I am a person with autism"

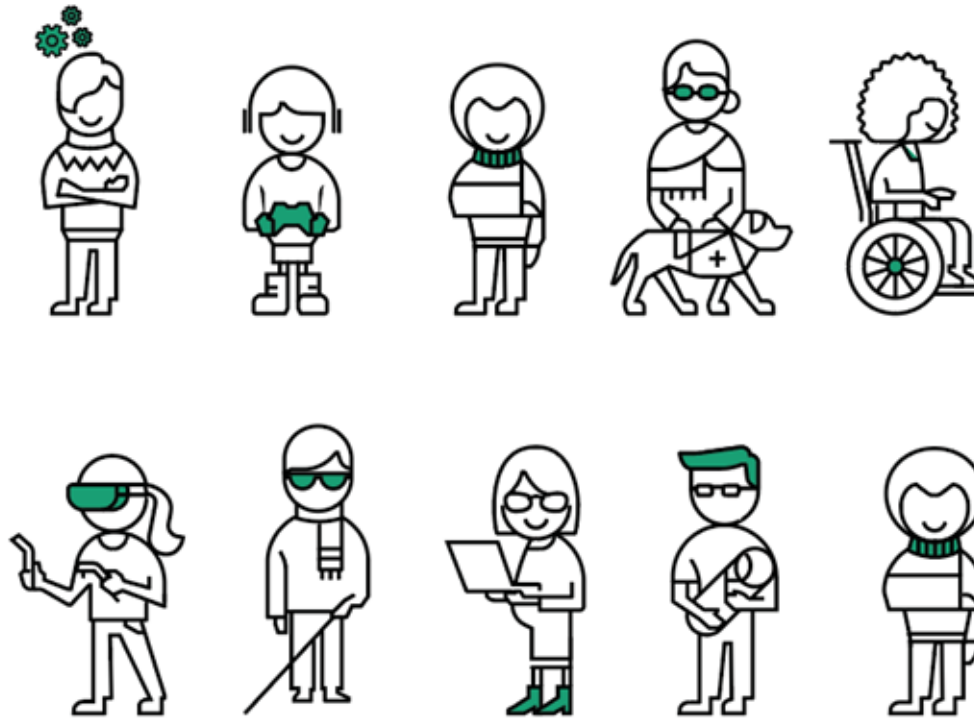
Identity-first language



"I am autistic"

Visible and Invisible Disabilities

- Some disabilities are easy to identify. Some are not.



Disabilities vary by time



Permanent



Temporary



Situational

Disabilities vary by time



Progressive



Dynamic

5 Basic Human Senses

1. Touch

- Touch, pressure, pain, vibration, texture, temperature

2. Smell

- Fragrant, woody, minty, sweet, chemical, toasted, lemon, fruity, pungent, decayed

3. Taste

- Sweet, sour, bitter, salt, spicy, umami

4. Visual

- Color, contrast, shape, form, movement

5. Sound

- Volume, pitch, distance, direction, rhythm

But wait, there are more!

6. Vestibular

- Balance and Movement

7. Proprioception

- Where your body is in space
- Smooth movement

8. Interoception

- Internal body and emotional states
- Hunger, fullness, thirst, urination, defecation, temperature, pain, heartbeat, tension, sadness, joy, anxiety

Human Motor Functionality

- Lower Body
 - Walking, Running, Jogging, Skipping, Kicking
 - Kneeling, Sitting, Thrusting, Jumping, Dancing
- Upper Body
 - Grasping, Reaching, Holding, Touching, Throwing
 - Using Tools, Twisting bottle caps, Doing up buttons, Using scissors, Brush hair, Brush teeth, Use utensils
 - Holding pencil, Drawing, Writing
 - One-handed, two independent hands, two coordinated hands
- Head
 - Speaking, Turning Head, Moving Eyes, Emoting

Kinds of Disability

1. See
 - Blind
 - Cataract
 - Distracted Driver
2. Hear
 - Deaf
 - Ear infection
 - Bartender
3. Speak
 - Mute
 - Laryngitis
 - Heavy accent
4. Move
 - Paralysis
 - Arm injury
 - New parent holding a baby
5. Think
 - Down's Syndrome
 - Concussion
 - Driving a car with the radio on
6. Chronic Illness
 - Muscular dystrophy
 - Heart disease
 - Seasonal affective disorder

Disabilities often co-occur

- 1 of 34 US children are diagnosed with Autism
 - 50-70% of autistic people also diagnosed with ADHD
- Deaf-blind (45,000 – 50,000 individuals in the USA)
- Parkinson's Disease (1 million people in USA)
 - Affects Mobility + Cognition + Speech

Activity: Disability Experiences

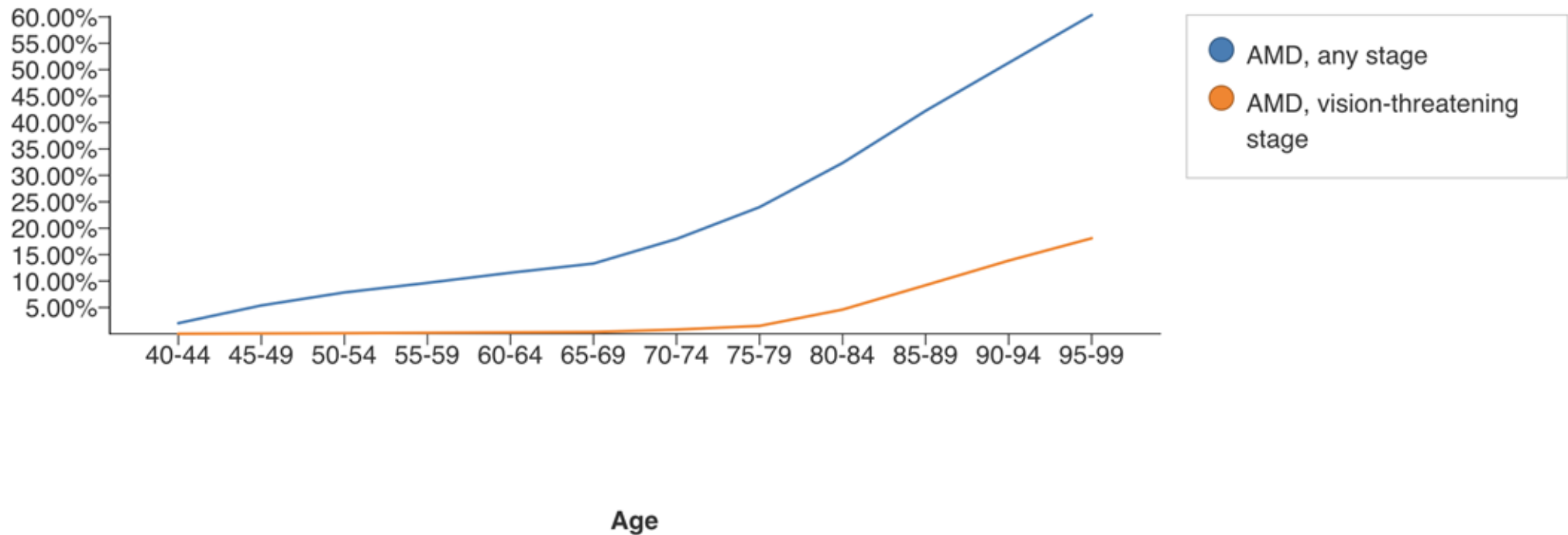
- Have you ever experienced a *situational* disability?
- Pair up with a neighbor and discuss your experience.
- Post a three-sentence writeup of your experience in our Slack **#in-class** channel.
 1. Set up the scenario.
 2. Describe the disabling experience.
 3. Explain how it was resolved.
- There will be a **PRIZE** for the most exciting experience.
- Tag your favorite post with a 🌟 emoji.

Vision Disabilities

- Labels: Blind, visually impaired, low-vision, color blind, night blind, loss of depth perception, lazy eye, four eyes
- Definitions
 - Visual acuity loss: $< 20/40$ (corrected)
 - Blindness: $< 20/200$ (corrected)
- Prevalence: 2% people in the USA (7 million)
 - 1 million are blind

Vision Loss Prevalence

Figure 1. Prevalence (Percentage of US Resident Population) with Vision Loss or Blindness (Best Corrected Visual Acuity 20/40 Or Worse), by Age Group (in years)



Causes of Vision Loss

- Permanent
 - Refractive errors (myopia, hyperopia)
 - Anophthalmia (born without eyes)
 - Retinoblastoma
- Temporary
 - Migraine
 - Keratitis
 - Conjunctivitis
 - Eye strain
 - Corneal abrasion
 - Detached retina
 - Stroke
- Situational
 - Sudden bright light or darkness
- Progressive
 - Diabetic retinopathy
 - Glaucoma
 - Age-related macular degeneration
 - Amblyopia (lazy eye)
 - Cataracts
 - Retinosis pigmentosa
 - Trachoma

Kinds of Vision Loss

What it might look like

Adapted from

<https://www.lighthouseguild.org>



BLURRY VISION
(CATARACT)



CENTER VISION LOSS
(MACULAR DEGENERATION)



SIDE VISION LOSS
(GLAUCOMA)



COMBINATION LOSS
(DIABETIC RETINOPATHY)



GLARE PROBLEM



CONTRAST PROBLEM

Famous People with Vision Impairments

Stevie Wonder



Claude Monet



Activity

- Who else can you find with a vision impairment?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with vision impairments.
- Post the pictures of your famous people in **#in-class**.

Hearing Disabilities

- Labels: Deaf, deaf, hard of hearing
- Definitions:
 - deaf: auditory sense channel is unavailable
 - hard of hearing: auditory sensing is limited (threshold > 20 dB)
 - ASL: American Sign Language – a visual language most used by people who are born deaf.
 - Deaf: people who are immersed in the (sign) language and culture of deafness
- Prevalence:
 - 0.25% of USA children born with hearing loss
 - 5% of all people on Earth have hearing loss (WHO)
 - 15% of USA adults over 18 have difficulty hearing.
- Hearing loss increases with age

Causes of Hearing Loss

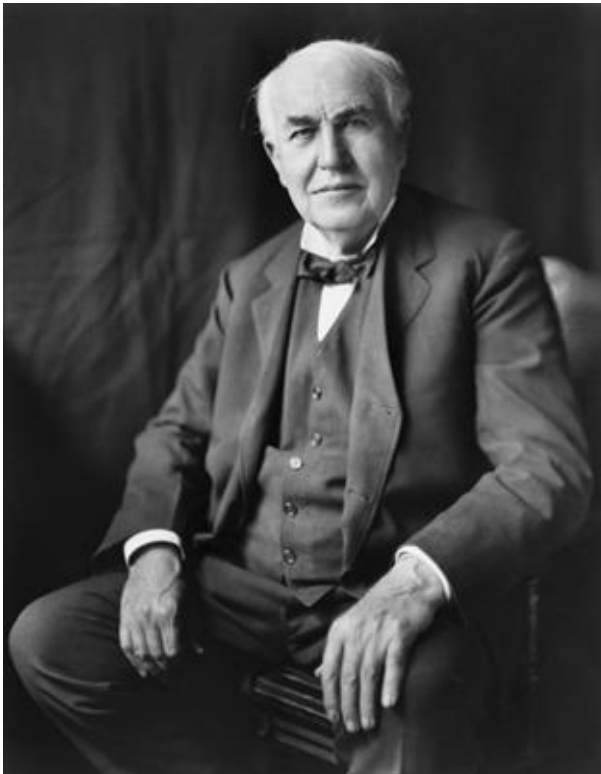
- Permanent
 - Damage to inner ear
 - Cancer
 - Tinnitus
 - Genetics
 - In utero (rubella, cytomegalovirus)
 - Birth asphyxia
 - Neurological
- Temporary
 - Earwax
 - Ear infection
 - Ruptured eardrum
- Situational
 - Attending a rock concert
 - Noisy restaurant
- Progressive
 - Otosclerosis
 - Age-related degeneration
 - Genetic hearing loss

Kinds of Hearing Loss

- Conductive: Sound prevented from passing through the outer ear or middle ear.
- Sensorineural: Damage to the inner ear
- Mixed: Both conductive and sensorineural
- Usually qualified as loss × pitch (frequency)
 - Mild
 - Severe
 - Profound

Famous People with Hearing Impairments

Thomas Edison



Nyle DiMarco



Activity

- Who else can you find with a hearing impairment?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with hearing impairments.
- Post the pictures of your famous people in **#in-class**.

Cognition

- Labels: Intellectual developmental disorder, autism, ADHD, dyslexia, dyscalculia, cerebral palsy, Tourette's, low IQ, dementia, learning disability
- There are many different kinds of disabilities here.

Prevalence of Cognition Disabilities in the USA

- IDD: 8.5%
- Autism: 3%
- ADHD: 9.4%
- Dyslexia: 15%
- Dyscalculia: 3-6%
- Dementia: 0.7%
- Tourette's: 0.6%
- Cerebral Palsy: 0.15%

Causes of Cognitive Disabilities

- Permanent
 - Genetics
 - Exposure to alcohol in utero
 - Infections during pregnancy
 - Lack of oxygen during birth
 - Traumatic brain injury
 - Stroke
- Temporary
 - Concussion
 - Drugs
- Situational
 - Distracted driving
- Progressive
 - Age
 - Heart disease
 - Parkinson's
 - Alzheimer's
 - Diabetes

Kinds of Cognitive Disability

- Practical Skills:
 - Taking care of oneself and handling everyday life skills
- Conceptual Skills:
 - Difficulty remembering things, using language, using numbers, making good decisions
- Social Skills:
 - Difficulty communicating with others

Famous People with Cognitive Differences

Temple Grandin



Greta Thunberg



Activity

- Who else can you find with a cognitive difference?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with cognitive differences.
- Post the pictures of your famous people in **#in-class**.

Speech

- Labels: Mute, Dysarthric speech, speech impediment, aphasia, stutterer, cleft palate, laryngitis
- Prevalence:
 - 5% of children have a speech impairment
 - 5% of children stutter for 6 months or more. 1% stutter long-term.

Causes of Speech Disabilities

- Permanent
 - Genetics
 - Traumatic brain injury
 - Dementia
 - Stroke
 - Dysarthria
 - Cancer
 - Hearing loss
- Temporary
 - Vocal cord damage
- Situational
 - Chewing
- Progressive
 - Huntington's
 - Parkinson's
 - ALS
- Dynamic
 - Selective Mutism
 - Shock/Surprise

Kinds of Speech Disability

- Disfluency: Altered flow of a person's speech
- Orofacial myofunctional: Using or moving your face, mouth, tongue muscles
- Speech sound: Ability to produce clear sounds
- Voice: Quality of your vocal sounds

Famous People with Speech Impediments

Ed Sheeran



Elvis



Activity

- Who else can you find with a speech disability?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with speech disabilities.
- Post the pictures of your famous people in **#in-class**.

Mobility or Motor

- Labels: Mobility impairment, Person who uses a wheelchair, prosthetic user, amputee, Motor Impairment
- Definitions: Motor skills involve motor, sensory, and cognitive abilities.
 - **Fine Motor Control (Dexterity)** - small muscle movements often the hands and fingers.
 - **Gross Motor Control (Mobility)** - typically refers to the larger muscle groups including the legs and impact walking and more generalized movement of the body
 - **Range of Motion** - required for different tasks may be vastly different

Prevalence of Mobility Disabilities

~12% of US Adults have a motor impairment that impacts their ability to walk or climb stairs

0.44% of developers according to 2022 Stack Overflow Developer Survey

Causes of Mobility Disabilities

- Permanent
 - Physical injury
 - Drug use in utero
 - Cerebral palsy
 - Traumatic brain injury
- Temporary
 - Broken limb
- Situational
 - New baby
- Progressive
 - Multiple sclerosis
 - Peripheral neuropathy
 - Repetitive stress injury
- Dynamic

Kinds of Mobility Disabilities

- Gross motor skills
- Fine motor skills
- Paralysis
- Limb Differences

Famous People with Mobility Impairments

Franklin D. Roosevelt



Judith Heumann



Activity

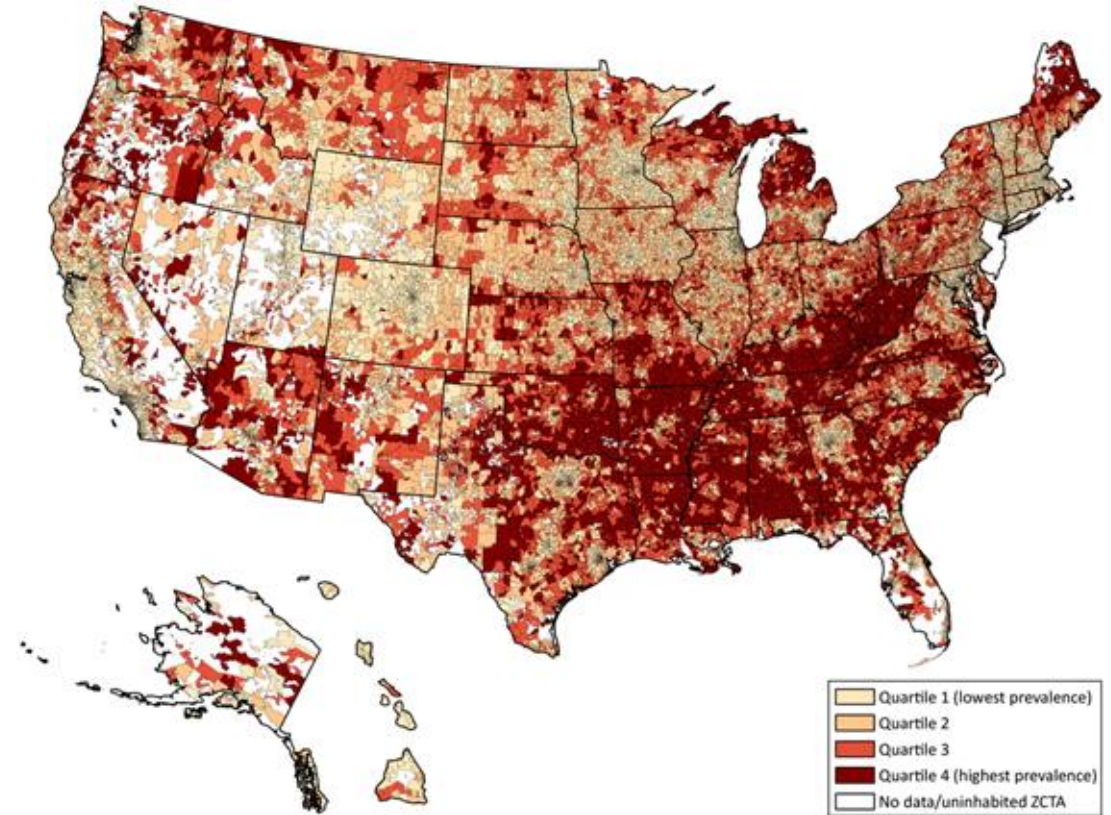
- Who else can you find with a mobility disability?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with mobility disabilities.
- Post the pictures of your famous people in **#in-class**.

Chronic Illness

- Labels: Chronically ill, chronic health conditions
- Definition:
 - Range of conditions and diagnoses that impact functioning and are not expected to go away or be immediately fatal.
 - Symptoms alter one's daily life.
- Prevalence:
 - 24.6% of US adults have 1 of 10 specific chronic conditions.
 - 27.2% of US adults have 2 or more of 10 specific chronic conditions.
 - Chronic conditions: hypertension, coronary heart disease, stroke, diabetes, cancer, arthritis, hepatitis, kidney failure, asthma, emphysema.

Prevalence of Chronic Illness

- Geographic distribution of chronic illness is *not* random.
- Chronic illness in the Southeast correlates strongly with decades of social and economics policies (e.g. low minimum wage, non-expansion of Medicaid), concentrated poverty, lack of access to affordable healthcare.



https://www.cdc.gov/pcd/issues/2024/23_0267.htm

Causes of Chronic Illness

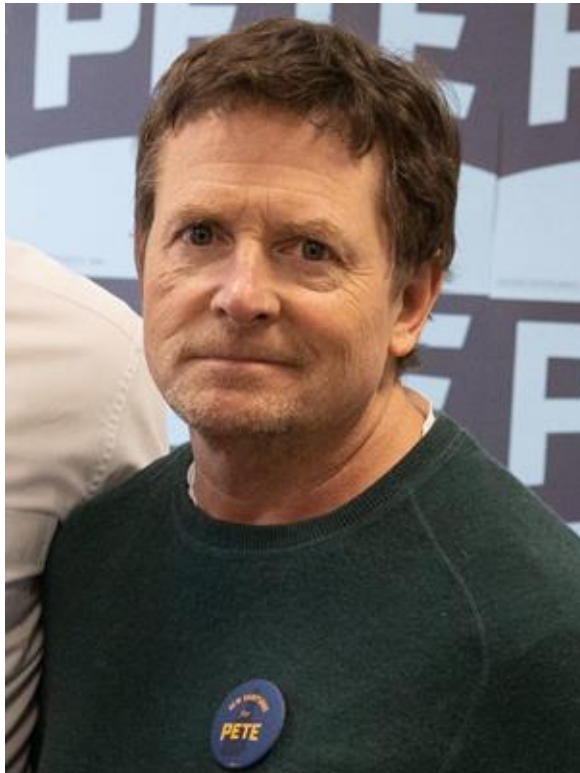
- Diseases
 - Parkinson's
 - ALS
 - Chronic Fatigue Syndrome
 - Multiple Sclerosis
- Life
 - Diet
 - Exercise
 - Stress
 - Mental Health
 - Access to healthcare

Attributes of Chronic Illness as a Disability

- Causes physical impairments
 - Pain, fatigue, discomfort, dizziness, nausea,
- Causes mental impairments
 - Disbelieved, cautious, vigilant about controlling symptoms,
- High variability in ability, often dynamic from day to day
- Impairments change after performing particular activities
 - Sometimes the choice to partake is worth the consequences.
 - e.g. eating a large dinner despite expected GI symptoms

Famous People with Chronic Illnesses

Michael J. Fox



Lady Gaga



Activity

- Who else can you find with a chronic illness?
- Use Google, ChatGPT, or other search engine to find pictures of 2 more famous people with chronic illnesses.
- Post the pictures of your famous people in **#in-class**.

Some co-occurring disabilities

- Deaf-Blind
- Parkinson's
- ALS
- AuDHD (Autism + ADHD)

Famous People with co-occurring disabilities

Helen Keller
Deaf-Blind



Haben Girma
Deaf-Blind



Famous People with co-occurring disabilities

Stephen Hawking
ALS



Lou Gherig
ALS



Famous People with co-occurring disabilities

Eminem
Autism + ADHD



Wentworth Miller
Autism + ADHD

